

Qualifying Exams Study Guide

Real, Complex Analysis; Linear Algebra, Group Theory, Galois Theory

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1 Analysis

1.1 Real

1.1.1 Fundamentals

1.1.2 Differentiation

Definition 1: Given $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$, the derivative at $x_0 \in \mathbb{R}^n$ is the linear map $Df(x_0) : \mathbb{R}^n \rightarrow \mathbb{R}^m$, if it exists, such that

$$\frac{\|f(x) - f(x_0) - Df(x_0)(x - x_0)\|}{\|x - x_0\|} \rightarrow 0 \text{ as } x \rightarrow x_0.$$

If we fix bases for $\mathbb{R}^n, \mathbb{R}^m$, then $Df(x_0)$ can be represented by a $m \times n$ matrix $J_f(x_0)$ called the *Jacobian* of f at x_0 .

Remark 1: When $m = 1$, one often writes $Df = (\nabla f)^\top$ where ∇f is viewed as a column vector in $\mathbb{R}^n$.

Proposition 1 (Properties of Derivative): Let $f, g : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $h : \mathbb{R}^k \rightarrow \mathbb{R}^n$, and $\alpha, \beta \in \mathbb{R}$.

- Df is unique when it exists
- $D(\alpha f + \beta g) = \alpha Df + \beta Dg$
- $D(fg) = (Df)g + f(Dg)$, when $m = 1$
- $D(f \circ h) = (Df)|_h \cdot (Dh)$ (*chain rule*)
- f differentiable at $x_0 \Rightarrow f$ (Lipschitz) continuous at x_0
- If Df exists at a point x_0 , f partially differentiable in all directions, and

$$J_f(x_0) = \left(\frac{\partial f_j}{\partial x_i}(x_0) \right)_{\substack{1 \leq i \leq n \\ 1 \leq j \leq m}}$$

- If all of the partial derivatives of f exist and are continuous at x_0 , then f differentiable at x_0

Remark 2: One need be careful with the last two properties - existence of partial derivatives needn't imply differentiability in general (we need some extra condition, such as continuity as stated here); consider

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}, \quad f(x, y) = \begin{cases} \frac{x^2 y}{x^2 + y^2} & (x, y) \neq 0 \\ 0 & (x, y) = 0 \end{cases}$$

One can check $\partial_x f, \partial_y f$ both exist (and equal 0) at the origin, but f not differentiable at 0. Intuitively, Df captures "average change in all directions" while partial derivatives capture "change in a specified direction".

Definition 2 (Other Derivatives): Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $F = (F_1, \dots, F_n) : \mathbb{R}^n \rightarrow \mathbb{R}^n$.

- $\frac{\partial}{\partial x_i} f(x_1, \dots, x_n) := \lim_{h \rightarrow 0} \frac{f(x_1, \dots, x_i + h, \dots, x_n) - f(x_1, \dots, x_n)}{h}$ is called the *partial derivative* of f in the x_i direction.
- Given a vector $d \in \mathbb{R}^n$ and $x \in \mathbb{R}^n$, define the *directional derivative* of f in the direction d at x by

$$df(x; d) := \lim_{h \rightarrow 0} \frac{f(x + hd) - f(x)}{h}.$$

- Define the *divergence* of F at x (assuming its first-partial derivatives exist) by

$$\operatorname{div} F(x) := \frac{\partial F_1}{\partial x_1}(x) + \dots + \frac{\partial F_n}{\partial x_n}(x).$$

- If $n = 3$, define the *curl* of F at x (assuming its first-partial derivatives exist) by

$$\operatorname{curl} F(x) := \det \left(\begin{pmatrix} e_1 & e_2 & e_3 \\ \partial_x & \partial_y & \partial_z \\ F_1(x) & F_2(x) & F_3(x) \end{pmatrix} \right) \\ = (\dots)$$

Proposition 2: If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ differentiable at $x \in \mathbb{R}^n$, then

$$df(x; d) = \nabla f(x)^T d.$$

If $d = e_i$ and f partially differentiable in x_i at x , then $df(x; d) = \frac{\partial f}{\partial x_i}(x)$.

Theorem 1 (Mean-Value Theorem): Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable on an open ball $B \subset \mathbb{R}^n$. Then for any $x, y \in B$, there exists a $z \in B$ such that

$$f(x) - f(y) = Df(z)(x - y).$$

PROOF. Define $\varphi(t) := f(y + t(x - y))$ for $t \in [0, 1]$, noting that $\varphi(0) = f(y)$ and $\varphi(1) = f(x)$. One computes, using the chain rule, that $\varphi'(t) = Df(y + t(x - y)) \cdot (x - y)$. By the one-variable mean-value theorem, there exists some $t_0 \in [0, 1]$ such that $\varphi(1) - \varphi(0) = \varphi'(t_0)$. Letting $z = y + t_0(x - y)$ gives the desired result, noting that z indeed in B by convexity.

For completeness, we also prove the one-variable mean-value theorem. ■

Theorem 2 (Clairaut's Theorem): Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$ twice differentiable at x_0 . Then, $\partial_{x_j} \partial_{x_i} f(x_0) = \partial_{x_i} \partial_{x_j} f(x_0)$ for all $i, j = 1, \dots, n$.

PROOF. (Sketch) For $n = 2$ (the proof extends to general dimensions by fixing all coordinates except the i th, j th) in variables (x, y) , consider the function

$$\varphi(s, t) := f(x_0 + s, y_0 + t) - f(x_0 + s, y_0) - f(x_0, y_0 + t) + f(x_0, y_0).$$

The idea is to recognize $\varphi(s, t)$ as a difference of functions in the s, t variables resp., apply mean-value theorem to pick-up $\frac{\partial}{\partial x}, \frac{\partial}{\partial y}$ terms, then use the definition of the second-order partial derivatives to estimate φ in terms of $\frac{\partial^2}{\partial x \partial y}, \frac{\partial^2}{\partial y \partial x}$. One should ultimately find that $\frac{\varphi(t, t)}{t^2} \rightarrow \frac{\partial^2 f}{\partial x \partial y}(x_0, y_0)$ and $\frac{\partial^2 f}{\partial y \partial x}$, depending on which variable one chooses to expand in first. ■

Theorem 3 (Inverse Function Theorem): Let $F : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}^n$ a C^k map such that $DF(x_0)$ invertible for some $x_0 \in \Omega$. Then there exist neighborhoods $U \subset \Omega$ of x_0 and $V \subset \mathbb{R}^n$ of $F(x_0)$ and a C^k map $G : V \rightarrow U$ such that

$$(F \circ G)(y) = y \quad \forall y \in V, \quad (G \circ F)(x) = x \quad \forall x \in U.$$

Moreover, $DF(x)$ invertible for $x \in U$, and

$$DG(y) = (DF)^{-1}(x), \quad y = F(x), x \in U.$$

Theorem 4 (Implicit Function Theorem): Let $F : \Omega := \Omega_x \times \Omega_y \subset \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^m$ be a C^k map, and assume $X_0 = (x_0, y_0) \in \Omega$ is such that $F(X_0) = 0$ and $D_y F(X_0)$ is invertible (where D_y the derivative in just the y -variables, i.e. represented by a $m \times m$ matrix). Then, there exists a neighborhood $U \subset \Omega_x$ of x_0 and a C^k function $h : \Omega_x \rightarrow \mathbb{R}^m$ such that

$$F(x, h(x)) = 0, \quad \forall x \in U.$$

PROOF. (Sketch) Define $f : \Omega \rightarrow \mathbb{R}^n \times \mathbb{R}^m$ by $f(X) := (X, F(X))$ ("augmenting" F to map to and from spaces of the same dimension). One checks

$$J_f = \begin{pmatrix} I_{n \times n} & 0_{n \times m} \\ D_x F & D_y F \end{pmatrix},$$

which has determinant $\det(D_y F)$ using properties of block matrices. This is non-zero at X_0 by assumption, so we may apply the inverse-function theorem, so locally there exists an inverse g . If we write $g = (\tilde{g}, \tilde{h}) : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n \times \mathbb{R}^m$, we see that $f(g(x, y)) = (x, y)$ (by inverse property) and $f(g(x, y)) = \tilde{g}(x, y), F(\tilde{g}(x, y), \tilde{h}(x, y))$ by definition. In particular we see that $\tilde{g}(x, y) \equiv x$, and thus we get the property

$$F(x, \tilde{h}(x, y)) = y.$$

Thus the map $h(x) := \tilde{h}(x, 0)$ gives the desired function. ■

Theorem 5 (Lagrange Multipliers, one constraint): Let $f, g : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}$ be C^1 and $\Sigma := \{x \in \Omega : g(x) = 0\}$, and assume $Dg(x)$ nonzero (i.e. the gradient of g) on all of Ω . Then, if f restricted to Σ has a max/min at $x_0 \in \Sigma$, then there exists $\lambda \in \mathbb{R}$ such that

$$\nabla f(x_0) = \lambda \nabla g(x_0).$$

PROOF. Let x_0 be a max/min of f restricted to Σ . Assume wlog that $\frac{\partial}{\partial x_n} g(x_0) \neq 0$. By implicit function theorem, then locally to x_0 , Σ is represented by $\{(x', h(x'))\}$ (where $x' \in \mathbb{R}^{n-1}$) for some C^1 function $h : \mathbb{R}^{n-1} \rightarrow \mathbb{R}$. Defining $F : \mathbb{R}^{n-1} \rightarrow \mathbb{R}$, $F(x') := f(x', h(x'))$, this implies $(x'_0, h(x'_0))$ a local max/min of F and thus its gradient must equal zero there. Chain rule gives

$$\nabla F(x') = \left(\frac{\partial}{\partial x'_i} f(x', h(x')) + \frac{\partial}{\partial x_n} f(x', h(x')) \frac{\partial h(x')}{\partial x_i} \right)_{i=1}^{n-1}. \quad (\dagger)$$

Also since $g(x', h(x')) = 0$ locally to x_0 , we can take the derivative of this expression and find

$$0 = \left(\frac{\partial g}{\partial x'_i}(x_0) + \frac{\partial h}{\partial x'_i}(x_0) \frac{\partial g}{\partial x_n}(x_0) \right)_{i=1}^{n-1}. \quad (\ddagger)$$

Taking $x' = x'_0$ in $(\dagger)$ makes the right-hand side 0. Equating $(\dagger)$, $(\ddagger)$ and simplifying like-terms yields the proof, where λ is found to be

$$\left[\frac{\partial f}{\partial x_n} \right] / \left[\frac{\partial g}{\partial x_n} \right]_{x_0}.$$

■

Remark 3: When we say “ Σ locally looks like V ”, what we really mean is there exists an open set U such that $\Sigma \cap U = V$.

Remark 4: In particular, then if f, g are C^1 on some open superset of Ω , and $\partial\Omega = \{x \in \Omega : g(x) = 0\}$, then one can compute global maxima of f by

- (i) computing *stationary points* i.e. such that $\nabla f = 0$, inside Ω , and
- (ii) computing points such that $\nabla f = \lambda \nabla g$, on $\partial\Omega$.

Then, if x_0 a global max of f , then either it lies inside Ω in which case $\nabla f(x_0) = 0$, or it lies on the boundary in which case there exists λ such that $\nabla f(x_0) = \lambda \nabla g(x_0)$. Thus, the global max of f is the point of greatest function value among those from 1., 2.. A similar statement holds for the global min.

Theorem 6 (Lagrange Multiplier, multiple constraints): Let $f, g_1, \dots, g_k : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}$ be C^1 . Let $\Sigma = \{x \in \Omega : g_i(x) = 0 \forall i = 1, \dots, k\}$, and assume $\{\nabla g_i(x)\}_{i=1}^k$ a linearly independent set of vectors for all $x \in \Sigma$. Then if f restricted to Σ has a local max/min at $x_0 \in \Sigma$, then there exists $\lambda_1, \dots, \lambda_k$ such that

$$\nabla f(x_0) = \lambda_1 \nabla g_1(x_0) + \dots + \lambda_k \nabla g_k(x_0).$$

Theorem 7 (Taylor's Theorem): Let $f : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}$ be a $C^{k+1}(\Omega)$ function. For any $x, x^0 \in \Omega$,

$$f(x) = f(x^0) + \sum_{j=1}^k \sum_{1 \leq i_1, \dots, i_j \leq n} \frac{\partial^j f}{\partial x_{i_1} \dots \partial x_{i_j}}(x_{i_1} - x_{i_1}^0) \dots (x_{i_j} - x_{i_j}^0) + R_k(x, x^0),$$

where the *remainder* R_k is such that

$$|R_k(x, x^0)| \leq M_k |x - x^0|^{k+1} \quad \text{as } x \rightarrow x^0,$$

where M_k depends only on the $(k+1)$ -st partial derivatives of f .

Remark 5: The easiest way to prove this is to apply 1-dimensional Taylor's theorem to the function $\varphi(t) := f\left(x^0 + t \frac{x - x^0}{\|x - x^0\|}\right)$, $t \in [0, \|x - x^0\|]$, for t sufficiently small, and expanding the derivatives of φ by chain rule.

On the other hand, the easiest way to derive the 1-diemnsional Taylor's theorem is by repeated application of the fundamental theorem of calculus,

$$\begin{aligned} \varphi(t) &= \varphi(t_0) + \int_{t_0}^t \underbrace{\varphi'(s)}_{\varphi'(t_0) + \int_{t_0}^s \varphi''(u) du} ds \\ &= \varphi(t_0) + \int_{t_0}^t \left[\varphi'(t_0) + \int_{t_0}^s \varphi''(u) du \right] ds \\ &= \varphi(t_0) + \varphi'(t_0)(t - t_0) + \int_{t_0}^t \int_{t_0}^s \varphi''(u) du ds, \end{aligned}$$

etc. In particular, one sees that this implies the remainder term can be written as

$$R_k(t) = \int_{t_0}^t \int_{t_0}^{s_1} \dots \int_{t_0}^{s_{k-1}} \varphi^{(k)}(u) du \dots ds_2 ds_1,$$

from which obtaining the bound on R_k follows by approximating this integral.

Theorem 8 (Taylor's Theorem, Weaker Assumptions): Let $f \in C^{k-1}(\Omega)$ and such that $D^{k-1}f$ is differentiable at $x^0 \in \Omega$. Then for $x \in \Omega$, the same conclusion as above holds, but with remainder

$$\lim_{x \rightarrow x^0} \frac{R_k(x)}{|x - x^0|^k} = 0,$$

i.e. we can't in general say anything about how *quickly* this quantity goes to zero.

Remark 6: In “asymptotic” notation, the first theorem says $R_k(x) = \mathcal{O}(|x - x^0|^{k+1})$ and the second says $R_k(x) = \mathcal{o}(|x - x^0|^k)$.

1.1.3 Vector Calculus

Definition 3 (Curve): A (parametrized) C^k curve in $\mathbb{R}^n$ is a connected set $\mathcal{C} \subset \mathbb{R}^n$ such that there exists a C^k function $\gamma : I \rightarrow \mathbb{R}^n$ with $\gamma(I) = \mathcal{C}$, where $I \subset \mathbb{R}$ an interval. We say $\mathcal{C}$ regular if γ' never vanishes on I .

A piecewise C^k curve is a set $\mathcal{C} = \mathcal{C}_1 \cup \dots \cup \mathcal{C}_m$ where each $\mathcal{C}_i$ a C^k curve and each has “shared endpoints” (in practice, think of something like a triangle).

Remark 7: More generally, a curve is generally defined as a connected subset of $\mathbb{R}^n$ that is “locally a parametrized curve”.

Definition 4 (Surface): A C^k surface in $\mathbb{R}^n$ is a connected set $S \subset \mathbb{R}^n$ such that for every point $p \in S$, there exists an open subset $U \subset \mathbb{R}^2$ and $V \subset \mathbb{R}^n$ such that $p \in V$, and a C^k homeomorphism $\varphi : U \rightarrow V \cap S$ such that $\varphi(U) = V \cap S$.

We call S regular if every such φ has $d\varphi_q : \mathbb{R}^2 \rightarrow \mathbb{R}^n$ is one-to-one for all $q \in U$, i.e. the Jacobian $J_\varphi(q)$ has rank 2.

Definition 5 (Tangent Plane): Let $S \subset \mathbb{R}^3$ a regular C^1 surface. The *tangent plane* at a point $p \in S$ is

$$T_p S = d_q \varphi(\mathbb{R}^2),$$

where $\varphi(q) = p$. Since $d_q \varphi$ one-to-one, this is a 2-dimensional subspace of $\mathbb{R}^3$.

Remark 8: This is well-defined, regardless of choice of parametrization, by an inverse function theorem argument.

Definition 6 (Normal): A normal to a regular C^1 surface $S \subset \mathbb{R}^3$ at a point p is a vector $n \in \mathbb{R}^3$ such that n is orthogonal to $T_p S$. If we restrict n to be of unit length, then since $T_p S$ two dimensional, there are precisely two such n 's, given by $\pm(v_1 \times v_2)$ where $\{v_1, v_2\}$ a basis for $T_p S$.

1.1.4 Integration

Definition 7 (Partition): Given a rectangle $R \subset \mathbb{R}^n$, a *box partition* of R is a **finite** collection $\mathcal{P} = \{B_k\}_{k=1}^N$ of boxes (rectangles) B_k such that $R = \bigcup_{k=1}^N B_k$.

Definition 8 (Upper/Lower Riemann Sum): Let $f : R \subset \mathbb{R}^n \rightarrow \mathbb{R}$ a bounded function. Given a partition $\mathcal{P}$ of R , define the *upper, lower Riemann sums*

$$U(f, \mathcal{P}) := \sum_{B \in \mathcal{P}} \sup_{x \in B} f(x) \cdot \text{vol}(B), \quad L(f, \mathcal{P}) := \sum_{B \in \mathcal{P}} \inf_{x \in B} f(x) \cdot \text{vol}(B),$$

where, if $B = [a_1, b_1] \times \dots \times [a_n, b_n]$, $\text{vol}(B) := (b_1 - a_1) \cdots (b_n - a_n)$. Define the *upper, lower Riemann integrals* of f by

$$\overline{\int_R f \, dx} := \inf_{\mathcal{P}: \text{partition of } R} U(f, \mathcal{P}), \quad \underline{\int_R f \, dx} := \sup_{\mathcal{P}: \text{partition of } R} L(f, \mathcal{P}).$$

In particular, we trivially have the inequalities

$$\inf_R f \cdot \text{vol}(R) \leq L(f, \mathcal{P}) \leq \underline{\int_R f \, dx} \leq \overline{\int_R f \, dx} \leq U(f, \mathcal{P}) \leq \sup_R f \cdot \text{vol}(R).$$

If the lower, upper Riemann sums agree, we write their shared value by $\int_R f \, dx$, called the Riemann integral of f over R .

Remark 9: We can extend this definition to general f over bounded domains Ω by taking R a rectangle such that $R \supset \Omega$ and extending f by zero outside of R .

Definition 9 (Content Zero): A set $\Omega \subset \mathbb{R}^n$ is said to be of *content zero* if for every $\varepsilon > 0$, there exists a finite number of boxes $B_1, \dots, B_N$ which cover Ω and such that $\sum_{n=1}^N \text{vol}(B_n) \leq \varepsilon$.

Theorem 9: If f bounded on R and has set of discontinuities with content 0, it is Riemann integrable on R .

Proposition 3: Riemann integrals are linear.

Theorem 10 (Fubini):

Definition 10 (Regular Partition):

Theorem 11 (Change of Variables):

1.1.5 Analysis on Functions

1.2 Complex

1.2.1 Analytic Functions

Definition 11 (Analytic/Holomorphic): A function $f : \mathbb{C} \rightarrow \mathbb{C}$ is said to be *analytic* on Ω if f is given by a converging power series everywhere in Ω . It is said to be *holomorphic* if f' exists.

For $z = x + iy \in \mathbb{C}$ and $f : \mathbb{C} \rightarrow \mathbb{C}$, write $f(z) = u(x, y) + v(x, y)i$ for real-valued functions u, v . Then, the *Cauchy-Riemann (CR) equations* are the equations

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

Theorem 12 (Cauchy-Riemann Equations): f is holomorphic if and only if u, v satisfy the CR equations.

Definition 12 (Contour Integration): Given a piecewise C^1 contour $C = C_1 \cup \dots \cup C_k$ with C_i parametrized by C^1 functions $\gamma_i : [t_i, t_{i+1}] \rightarrow \mathbb{C}$, define

$$\int_{C^\pm} f(z) dz := \pm \sum_{i=1}^k \int_{t_i}^{t_{i+1}} (f \circ \gamma_i)(s) \cdot \gamma_i'(s) ds,$$

where the $\pm$ indicates the *orientation* of C .

A domain Ω is said to be *simply connected* if for any two curves $\gamma_0, \gamma_1 : [0, 1] \rightarrow \mathbb{C}$ with common endpoints α, β in Ω , there exists a function $\gamma_t : I \rightarrow \mathbb{C}$ for $t \in [0, 1]$ such that $(t, s) \mapsto \gamma_t(s)$ is a continuous map, $\gamma_0 = \gamma_0$ and $\gamma_1 = \gamma_1$, and $\gamma_t(0) = \alpha, \gamma_t(1) = \beta$ for all $t \in [0, 1]$.

A curve C is said to be *simple* if it does not intersect itself, i.e. it has an injective parametrization, and *closed* if its parametrization $\gamma : I \rightarrow \mathbb{C}$ is equal on the endpoints of I .

Theorem 13 (Cauchy): Let f be a holomorphic function defined on Ω being simply connected with C a simple closed curve contained in Ω . Then

$$\int_C f(x) dz = 0.$$

Theorem 14 (Cauchy's Integral Formula): Let f holomorphic on Ω , which contains a circle C and its interior. Then

$$f(z) = \int_{C^+} \frac{f(w)}{w - z} dw, \quad \forall z \in \Omega.$$

In particular, this implies f is analytic on all of Ω , with

$$f^{(n)}(z) = \frac{n!}{2\pi i} \int_{C^+} \frac{f(w)}{(w - z)^{n+1}} dw,$$

and thus

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n,$$

for any $z_0 \in \Omega$ and z such that $z - z_0 \in \Omega$.

Corollary 1 (Cauchy's Inequalities): Let $f : \Omega \rightarrow \mathbb{C}$ holomorphic. Then

$$|f^{(n)}(z)| \leq \frac{n! \|f\|_{C_R(z), \infty}}{R^n},$$

where $R > 0$ any real number such that $D_R(z) \subset \mathbb{C}$, where $D_R(z)$ the R -radius disc centered at z , and

$$\|f\|_{C_R(z), \infty} := \sup_{w \in D_R(z)} |f(w)|.$$

Theorem 15 (Argument Principle): Let f be a meromorphic function on the interior of a simple closed curve $C \subset \mathbb{C}$, such that f has no roots nor poles on C . Then,

$$\frac{1}{2\pi i} \int_C \frac{f'(z)}{f(z)} dz = \#\{\text{zeroes of } f \text{ inside } C\} - \#\{\text{poles of } f \text{ inside } C\},$$

the right-hand side counted with multiplicity.

Theorem 16 (Rouché's): Let f, g be holomorphic functions on Ω with a curve $C \subset \Omega$ such that

- $|f| > |g|$ on C ,
- both f, g never vanish on C .

Then, f and $f + g$ have the same number of zeros in the interior of C .

1.2.2 Harmonic Functions

1.2.3 Conformal Mappings

2 Algebra

2.1 Linear

2.1.1 Elementary

2.1.2 Vector Spaces

2.1.3 Diagonalization and Related

2.2 Groups

Definition 13 (Groups, Action): A group G is a set equipped with a binary operation $\cdot$ satisfying the axioms:

- $f \cdot (g \cdot h) = (f \cdot g) \cdot h, \forall f, g, h \in G$
- $\exists 1 \in G$ s.t. $g \cdot 1 = 1 \cdot g = g \forall g \in G$
- $\forall g \in G, \exists g^{-1} \in G$ s.t. $g \cdot g^{-1} = g^{-1} \cdot g = 1$

A group action on a set X is a function $\varphi : G \times X \rightarrow X$ satisfying:

- $\varphi(h, \varphi(g, x)) = \varphi(h \cdot g, x), \forall g, h \in G, x \in X$
- $\varphi(1, x) = x \forall x \in X$

When clear from context, we write $g \cdot x = \varphi(g, x)$. When $\varphi(G, X) := \{\varphi(g, x) : g \in G, x \in X\}$ is equal to X , we say the action of G is *transitive*. A set X equipped with a group action of G is sometimes called a G -set.

Definition 14 (Homomorphism): A group homomorphism $\varphi : G_1 \rightarrow G_2$ is a function such that

$$\varphi(gh) = \varphi(g)\varphi(h)$$

for all $g, h \in G_1$. When φ a bijection we call it a group isomorphism.

Definition 15 (Permutation, Symmetric, Alternatic Groups): Let X a finite set of n elements. The *symmetric group* of X is the set of bijections of X , i.e.

$$S_X := \{f : X \rightarrow X \mid f \text{ a bijection}\}.$$

When $X = \{1, \dots, n\}$, we often write $S_n \equiv S_X$. We call $f \in S_X$ a *permutation* of X .

The *sign*, or *sgn*, of a permutation $f \in S_n$ is defined to be

$$\text{sgn}(f) := (-1)^{N(f)}, \quad N(f) := \#\{(x, y) \in X \times X \mid x < y \text{ and } f(x) > f(y)\},$$

i.e. the number of “inversions” that f contains. We sometimes say f is even/odd if $\text{sgn}(f) = +1/-1$.

The *alternating group* on n elements is the subgroup

$$A_n := \{f \in S_n \mid \text{sgn}(f) = 1\}.$$

Proposition 4:

- $\#S_n = n!, \#A_n = \frac{n!}{2}$
- a *transposition* of $\{1, \dots, n\}$ is a permutation that only interchanges two elements and fixes the rest. Any permutation can be written as a composition of transpositions. Moreover, the sign of a permutation is equal to the parity of (any) number of transpositions needed to write the permutation

Definition 16 (Orbit, Stabilizer): Let G act on a set X . Define, for $x \in X$,

$$\mathcal{O}_x := \text{orbit of } x \text{ under } G = \{g \cdot x : g \in G\} \subset X,$$

$$G_x := \text{stabilizer of } x \text{ under } G = \{g \in G : g \cdot x = x\} \subset G.$$

Remark 10: The orbit is “everywhere x can go” (under G), and the stablizer is what “fixes” x (under G).

Definition 17: A *subgroup* H of a group G is a subset of G which is still a group when endowed with the operation from G (so in particular it is closed under the operation). A (*left*) *coset* of H in G is a set of the form

$$aH = \{ah : h \in H\},$$

where a some element of G . One denotes $G/H = \{\text{set of cosets of } H\}$. Equivalently, there is a natural action of H acting on G as a set (by right-multiplication), given by $h \star g := g \cdot h$; thus cosets of H are just orbits under this action.

Theorem 17 (Lagrange): Let $H \subset G$ a finite subgroup of a finite group. Then all cosets of H have the same cardinality, and in particular are disjoint. In particular, we have

$$|G| = |G/H| \cdot |H|,$$

and so $|H|$ divides $|G|$.

PROOF. The disjointness follows from the “orbit of an action” characterization (namely, orbits are either disjoint or equal). For the equality of cardinality, fix $a \notin H$ and define the map $f : H \rightarrow aH, h \mapsto ah$. This is an injection, since $f(h_1) = f(h_2) \Rightarrow ah_1 = ah_2 \Rightarrow h_1 = h_2$ by multiplying both sides by a^{-1} . This is also a surjection since given $ah \in aH$, one simply maps h by f . This shows $|aH| = |H|$ and since a was arbitrary, completes the proof. The “in particular” follows from the partitioning result. ■

Proposition 5: Let X a transitive G -set. Then there exists a subgroup H of G such that X is isomorphic (as a G -set) to $X \cong G/H$, equipped with the left-action

$$(g, aH) \in G \times G/H \mapsto g \cdot aH := (ga)H \in G/H.$$

Thus, $|X|$ divides $|G|$ and the *orbit-stabilizer formula* holds:

$$|G| = |X| \cdot |G_x|, \quad \forall x \in X.$$

Remark 11: When we say “ X_1, X_2 isomorphic as G -sets”, we mean a bijection which respects the group actions, i.e. $f : X_1 \rightarrow X_2, f(g \cdot_1 x) = g \cdot_2 f(x)$.

Remark 12: If X not transitive, we can reword this proof by taking any orbit $\mathcal{O}_x$ in X , in which case the formula reads $|G| = |\mathcal{O}_x| \cdot |G_x|$.

PROOF. Fix $x \in X$ and let $H := G_x$, and let $f : G/H \rightarrow X$ be given by $f(aH) := a \cdot x$. ■

Definition 18 (Normal subgroup, quotients): Given a group G , a subgroup $H \subset G$ is said to be *normal* if it is closed under conjugation by G , that is,

$$ghg^{-1} \in H, \quad \forall g \in G, h \in H.$$

The *quotient group* is the group G/H of cosets of G with respect to H equipped with left-multiplication (one checks that this is indeed a well-defined group when H normal in G).

G is said to be *solvable* if there exists a sequence of subgroups

$$\{1\} = G_0 \subset G_1 \subset \dots \subset G_{n-1} \subset G_n \subset G$$

such that G_{i-1} a normal subgroup of G_i and G_i/G_{i-1} abelian, for $i = 1, \dots, n$.

Proposition 6: Let $\varphi : G \rightarrow H$ a group homomorphism. Then $N := \ker(\varphi)$ a normal subgroup of G .

Moreover, φ induces an injective homomorphism $\tilde{\varphi} : G/N \rightarrow H$ defined by $\tilde{\varphi}(aN) := \varphi(a)$. In particular, $\text{im}(\varphi)$ is isomorphic to G/N .

PROOF. If $h \in N, g \in G$, then $\varphi(ghg^{-1}) = \varphi(g)\varphi(h)\varphi(g)^{-1} = \varphi(g) \cdot \varphi(g)^{-1} = 1$ so $ghg^{-1} \in N$.

We need to show $\tilde{\varphi}$ well-defined. If $aN = a'N$, then there exists $h \in N$ such that $a = a'h$. So

$$\tilde{\varphi}(aN) = \varphi(a) = \varphi(a'h) = \varphi(a')\varphi(h) = \tilde{\varphi}(a'N) \cdot 1 = \tilde{\varphi}(a'N),$$

so the function is well-defined. It is injective since

$$\tilde{\varphi}(aN) = 1 \Leftrightarrow \varphi(a) = 1 \Leftrightarrow a \in N \Leftrightarrow aN = N,$$

but $N = 1_{G/N}$, so φ injective.

■

2.3 Rings

2.4 Fields & Galois Theory